PM1 Exercise

Example 1: Consider a ~~Cortex-M4~~ processor which has a ~~3-stage~~ 5-stage instruction pipeline running code in a loop. Assume:

- There are ~~4~~ 6 instructions to run the basic task in a loop, say copying data from one memory location to another.
- There ~~is one~~ are two additional branch instructions at the end of the loop go back to the beginning of the loop to form the loop.
- When the branch instruction is executed, the pipeline is flushed, and we need to start over the pipeline.

Question:

1. What is the throughput of this loop?
2. If we modify the loop to repeat the ~~4~~ 6 instructions ~~5~~ 4 times in the loop, what is the throughput of the new longer loop?

Solution:

1. Including the branch instruction, there are 8 instructions in the loop~~, as shown in the ASM pseudo below~~. So the total number of clock cycles for running the loop is 8+4 = 12. Hence, the throughput is (8 Insts)/(12 cycs) = $\frac{2}{3}$ Insts/c.

2. If we modify the loop to repeat the ~~4~~ 6 instructions ~~5~~ 4 times in the loop, as shown in the ASM pseudo below, there will be 6×4=24 instructions for the basic task and 2 instruction for the branch. Altogether, we have 26 instructions, which takes 26+4=30 clock cycles. Then the throughput is (26 Insts)/(30 cycs) = $\frac{13}{15}$ Insts/c.

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